PORTFOLIO DISSERTATION 2026

Dr Adheesh Budree
a.budree@qub.ac.uk

01/06/2026

AGENDA

Topic	Time
Dissertation Overview	10-1
Break	1-2
Data Sources	2-3
Q&A	3-4

QUEEN'S UNIVERSITY BELFAST

ANNOUNCEMENTS

Datathon Information

- 11th June – focuses on data analytics training
- 12th Main event
  - Company mentors
  - Food is provided
  - Held in the Student Hub
  - Prizes
  - T-shirt
  - You should bring your own laptops (Although the lab has also been booked)

QUEEN'S UNIVERSITY BELFAST

LICENSES

Alteryx register at: https://www.alteryx.com/sparked

Tableau register at: https://www.tableau.com/university-students

Also remember you have access to **datacamp** for courses you might want to do over the semester: https://www.datacamp.com/

DISSERTATION WORKSHOP

QUEEN'S UNIVERSITY BELFAST

MSc Business Analytics - Dissertation Portfolio Timetable 2026 – GIFT City

Element	Date	Time	Location	Person
Dissertation training day 1: Dissertation overview and requirements	01/06/2026	10am – 4pm	Lecture Theatre	Adheesh
Dissertation training day 2: Data sources and ML Models Recap	02/06/2026	10am – 4pm	Lecture Theatre	Shriya
Dissertation training day 3: Interpretable ML and Deep Learning	03/06/2026	10am – 4pm	Lecture Theatre	Shriya
Dissertation training day 4: Drop-in session	04/06/2026	12pm – 2pm	Lecture Theatre / MS Teams	
Topic Overview Form Required	**05/06/2026**	**4pm**	**Upload to canvas**	
Datathon	11/06/2026	2pm - 5pm	GIFT City Building	
Datathon	12/06/2026	8.30am – 6:30pm	GIFT City Building	
Project proposal/feasibility study deadline (25%)	**03/07/2026**	**4pm**	**Upload to canvas**	
Dissertation workshop 1: A. Data science research methodology B. State of the art machine learning - I	06/07/2026	9am-12noon	Lecture Theatre	Shreya
Dissertation workshop 2: C. State of the art machine learning - II	13/07/2026	10am-4pm	Lecture Theatre	Shreya
Dissertation workshop 3: D. Results, discussion and managerial implications	20/07/2026	1pm-4pm	Lecture Theatre	Shreya
Dissertation portfolio deadline (75%)	**11/09/2025**	**4pm**	**Upload to canvas**	

QUEEN'S UNIVERSITY BELFAST

Semester 2	**Learning outcomes**
Research Proposal 1	Explain the purpose and structure of a research proposal; write aims, objectives, and research questions
Research Proposal 2	Explain the purpose and structure of a research proposal; write aims, objectives, and research questions
Writing a literature review	Synthesise and critique relevant academic sources to produce a cohesive and insightful literature review.
Writing a Methodology	Structure and explain a methodology using appropriate language; link findings to literature and cite correctly.
Writing a results & discussion section	Analyse and interpret findings; integrate relevant literature and cite appropriately.
Writing Introductions and Conclusions and abstract	Write effective introductions, conclusions, and abstracts with appropriate content and language
Reflective Writing	Explain the purpose of a reflective writing piece and describe and implement common grammatical and lexical patterns in reflective writing
Finalising the dissertation and tutorial sessions	Make final edits, proofread and format a dissertation in line with guidelines from the school.

QUEEN'S UNIVERSITY BELFAST

RESOURCES

- General R coding/data science: https://r4ds.hadley.nz/ or https://adv-r.hadley.nz/
- caret: https://topepo.github.io/caret/
- Interpretable Machine Learning (IML): https://christophm.github.io/interpretable-ml-book/

RESITS AND PROGRESSION

- If you have failed any S1 modules or have issues around progression you will be contacted after the June exam board
- Students need to pass all modules
- Students can proceed to the dissertation with one failed module, which MUST be passed before the dissertation is submitted
- Students with more than one fail need to pass these modules before proceeding to the dissertation

QUEEN'S UNIVERSITY BELFAST

IMPORTANT DATES

Topic: Friday 5th June

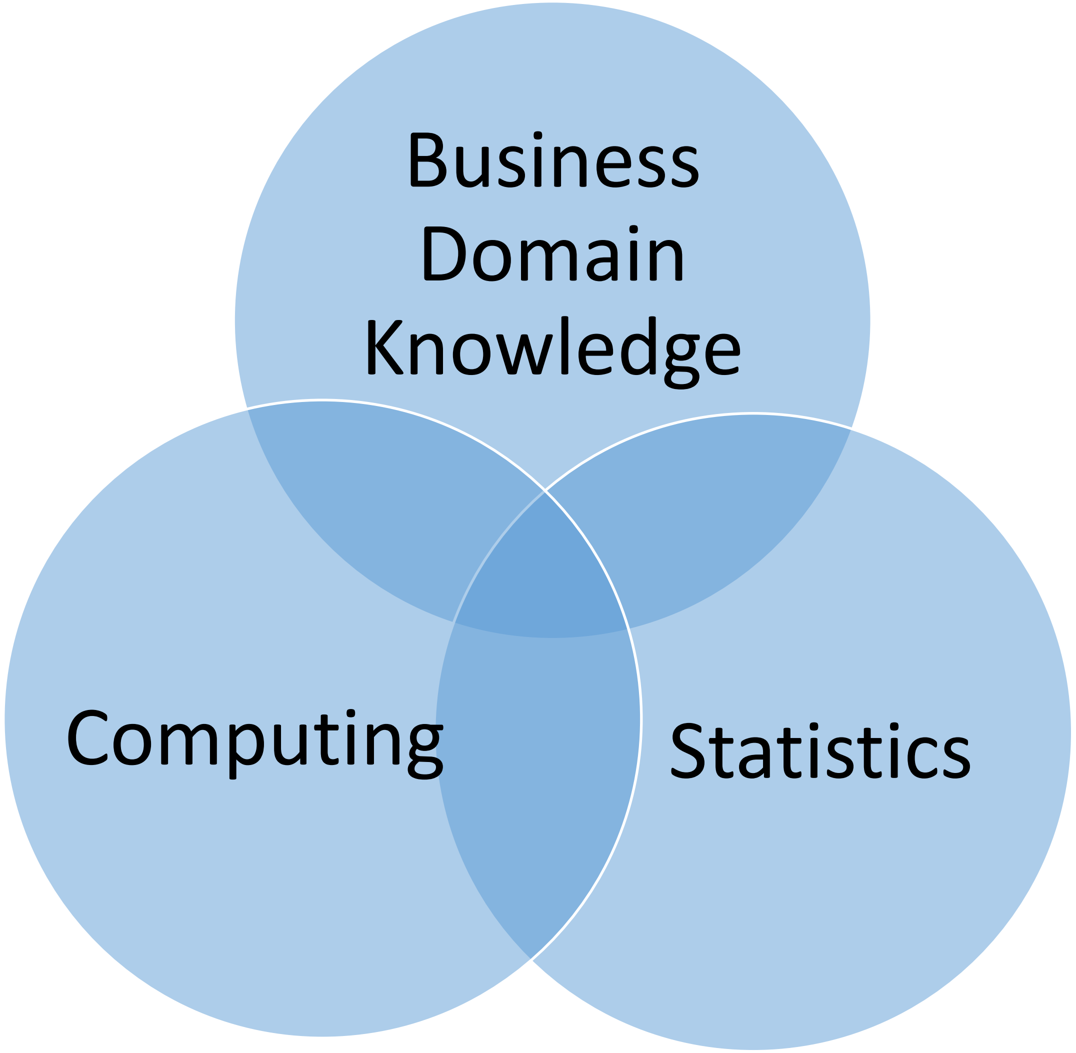
Start: Mid June (after semester 1 exam board)

Full Proposal and Feasibility Study: 3rd July

Submission: 11th September 2026 (before 4pm)

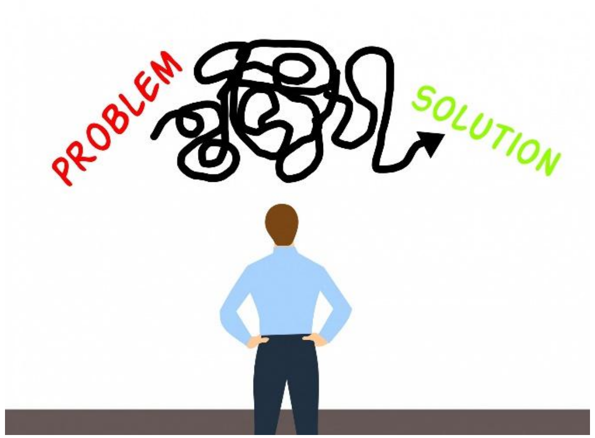
QUEEN'S UNIVERSITY BELFAST

MSc BUSINESS ANALYTICS STRUCTURE: CORE KNOWLEDGE BASE



QUEEN'S UNIVERSITY BELFAST

MSc BUSINESS ANALYTICS STRUCTURE: SOLUTION COMPONENTS



Data Acquisition and Storage

Prescriptive

Predictive

Descriptive

QUEEN'S UNIVERSITY BELFAST

MSc BUSINESS ANALYTICS STRUCTURE: MODULES

Semester 1	Semester 2	Semester 3
Advanced Analytics and Machine Learning	Dissertation Project	Statistics for Business
Data Mining		Data Management
Marketing Analytics		HR Analytics
Data Driven Decision Making		AI in Business and Society

DISSERTATION LEARNING OBJECTIVES

On completion of the dissertation, students will be able to:

- Undertake and manage a small business analytics project.
- Develop a business analytics solution.
- Critically evaluate the role of the solution in solving a specific problem, and in particular the strengths and limitations of the solution in solving the problem.
- Evaluate the legal and ethical implications of the solution.
- Synthesise, analyse, interpret, and critically evaluate information from a variety of different sources.

DISSERTATION OVERVIEW

- You can choose your own topic (within certain parameters).
- Most students use secondary data - what areas you are most interested in; review potential data sources to see what data is available.
- You cannot use data that has been used in a previous assessment.
- All dissertations must contain two general elements:
  - Technical component: normally involving the techniques used in the course (e.g. machine learning)
  - Written component

DISSERTATION PORTFOLIO

- 60 CATS
- 3 LINKED ELEMENTS:
  - 1. A project proposal / feasibility study of 2500 words + code in appendices (25%)
    - Due 03/07/2026 4pm
  - 2. A research report of 7500 words detailing the research that was carried out (50%).
    - Due 11/09/2026 4pm
  - 3. A technical report, logbook and reflective discussion of 2500 words + (code in appendices) (25%);
    - Due 11/09/2026 4pm

QUEEN'S UNIVERSITY BELFAST

WORD LENGTH

- The word count for the project proposal and feasibility study is 2500 words.
- The word count for the research report is 7500 words.
- The word count for the technical report, logbook, and reflective discussion is 2500 words.
- The word count includes the title page, acknowledgements, contents, tables and footnotes, but excludes appendices and the reference list. The word count should be clearly stated on the front cover of each submission element.
- It is permissible to exceed the word limit by 10% only. Once the word limit has been **exceeded by more than 10%, a penalty of 10 marks deducted from the mark awarded will be imposed.**
- Students who do not include a word count on the submission will normally be penalised by the deduction of 10 marks from the mark awarded.

QUEEN'S UNIVERSITY BELFAST

Important Information on Grading

- The overall mark recorded for the module will be out of 100%
- Each element will be marked out of 100 drawing from the PGT Conceptual Equivalent Framework. The mark will then be adjusted to 25%, 50% and 25% respectively for each of the three elements, before being summed up for a total mark out of 100% for the entire 60 CAT module.
- All three elements will be marked after the final submission deadline in September.
- Students must attain an aggregate mark of 50% or above to pass the module.
- Students must achieve an overall mark of 50% or higher to be considered for the award of Masters in Business Analytics.
- A sample of dissertations are moderated by the internal and external examiner to ensure marking is fair and consistent.

QUEEN'S UNIVERSITY BELFAST

GRADING

7.1.4 For Postgraduate Certificate, Postgraduate Diploma and Master's Degree results there is a common mark scale as follows:

70+	Pass with distinction*
60+	Pass with commendation
50+	Pass
Below 50	Fail

*For Master's Degrees, a pass with distinction will be awarded only when the following three conditions have been satisfied: an overall programme mark of 70+ is achieved and a mark of 70+ is achieved in the dissertation module and a weighted average (truncated to an integer) of 65+ is achieved in the other modules.

These mark scales must be applied by all Boards of Examiners except where the Director of Education and Student Services has, following application from the School, granted exemption from their use.

https://www.qub.ac.uk/directorates/AcademicStudentAffairs/AcademicAffairs/GeneralRegulations/StudyRegulations/StudyRegulationsforPostgraduateTaughtProgrammes/

QUEEN'S UNIVERSITY BELFAST

SUPERVISION

- 2 Meetings
- Supervisor can review one chapter
- Cannot switch supervisor (as all are at maximum capacity)

QUEEN'S UNIVERSITY BELFAST

CHOOSING A TOPIC

INITIAL TOPIC FORM

- See canvas for template
- Please submit by Friday 5th June

THINGS TO CONSIDER WHEN CHOOSING A TOPIC

DATA

INTERESTS

CAREER

SKILLS

QUEEN'S UNIVERSITY BELFAST

PROPOSAL AND FEASIBILITY STUDY

PROPOSAL AND FEASIBILITY STUDY

25%, 2500 Words

Suggested structure:

- Introduction and aims
- Background
- Proposed Methodology
- Data quality assessment
- Preliminary data analysis (basic summary stats / visuals / data quality).
- This is to ensure the project is technically possible.
- Conclusion
- Project plan
- Appendix 1: Completed ethical approval form
- Appendix 2: Any code used

QUEEN'S UNIVERSITY BELFAST

INITIAL LITERATURE SEARCH

- Pick the five most important papers and summarise these. The review will then be expanded out in the subsequent research report.
- Two areas:
  - Problem/contextual literature
  - Technical literature (i.e. what methods have other studies used to address the problem)
- Often efficient to search first on google scholar
  - Keyword search
  - Cited by
  - Related articles
- Proceeding to specialist databases (sciencedirect, proquest, Elsevier etc.)

QUEEN'S UNIVERSITY BELFAST

(PROPOSED) METHODOLOGY

Two elements should be considered:

1. Discuss the methodology used to carry out the descriptive analysis / data quality etc. in the feasibility study
2. Discuss the proposed methodology/plan for the remainder of the project – at this stage this can be a summary of the main points, as this will be developed more fully for the research report and technical submission.

- CRISP-DM
- Summarise the main proposed steps in the methodology.
  - consider: the data, what analytics techniques you will use, what tools you will use.
- Consider any ethical / legal issues (N.B. the ethics form needs to be included as an appendix – this is also assessed).

QUEEN'S UNIVERSITY BELFAST

PRELIMINARY ANALYSIS

The aim is to ensure the project is feasible e.g. is the data of high enough quality, is there enough data, are there any issues that weren't originally apparent?

Tasks:

- Data quality assessment
- Numerical summaries
- Visualisations
- Correlation / Association

N.B. you do not have to carry out any prediction / regression for the preliminary analysis (in the example projects you may see this, but the requirements have changed slightly from last year).

QUEEN'S UNIVERSITY BELFAST

ETHICAL CONSIDERATIONS AND APPROVAL

- See canvas / appendix 2 of the dissertation handbook

PROPOSAL AND FEASIBILITY STUDY GRADING (HANDBOOK, APPENDIX 7)

- Conceptual Equivalents Scale
  - Used to evaluate all written work

- Marking Rubric

SAMPLE PROJECT A: FORECASTING VEHICLE CO2 EMISSIONS

1. Primary question:

Which model has the highest accuracy in predicting CO2 emissions using different machine learning algorithms?

2. Sub-questions:

- Which factors are most important for predicting CO2 emissions?
- How does the choice of fuel type impact the level of CO2 emissions in vehicles?
- How do business policies and incentives for renewable energy adoption impact CO2 emissions reduction targets and the overall sustainability of businesses?

- **Literature: why reducing carbon emissions is an important topic; past literature using ML to solve this problem.**
- **Proposed Methodology:** CRISP-DM, Overview of the proposed methodology.
- **Data:** [from Kaggle](), 26k observations and 15 variables. Target variable is CO2 emissions from the car.
- **Analysis:** Addressing data quality issues, Data visualisation, correlation analysis, regression (n.b. regression not required in this years feasibility study); IML (also not required for the feasibility study this year).
- **Project plan:** Detail on tasks including a GANTT chart, identification of any project risks.
- **Ethical approval form**
- **Code in Appendix**

N.B. there are some changes to the feasibility study requirements this year:

- literature review to focus on 5 most important papers
- preliminary analysis should focus on descriptive analysis, rather than prediction (i.e. you don't need to do regression for the feasibility study).
- Some changes to the content of the methodology section

QUEEN'S UNIVERSITY BELFAST

RESEARCH REPORT

QUEEN'S UNIVERSITY BELFAST

RESEARCH REPORT

- Title page
- Declaration
- Acknowledgements (optional)
- Abstract
- Table of Contents
- Introduction
- Background / Literature
- Methodology
- Findings
- Discussion of Findings
- Conclusions
- Recommendations
- Reference list

- 7500 words
- 50%
- n.b. The structure is very similar to a typical journal style paper. Detailed technical information / code should be included in the technical report.

QUEEN'S UNIVERSITY BELFAST

INTRODUCTION AND BACKGROUND

Introduction - This section should explain the context within which the report is situated, as well as the business/research problem. You should include the research question, aims and objectives of the project, and the intended contribution to practice and to the wider body of knowledge of the topic. The section should include an outline of the report structure.

Background – In this section you should provide a summary of the research and practitioner literature that is most relevant to the topic. For example, you should consider prior approaches to solving the problem, and relevant theory or research that is relevant to the topic. You should use relevant keywords to search online databases such as Google Scholar, Science Direct, IEEE Xplore, and Business Source Premier. Based on this literature, you should summarise the 'state of the art' in your chosen topic.

QUEEN'S UNIVERSITY BELFAST

METHODOLOGY

Methodology – The methodology is of crucial importance to the success of a business analytics report, and, as such, this is a critical section in the report.

You should, therefore, explain in detail the design of the project, the data used, and the methods, tools and techniques employed to store, process and analyse the data, so that the reader can evaluate the validity of your findings, conclusions and recommendations. Below is a possible format for this section-

- Restate the problem / questions you are investigating and the project aims and objectives.
- Discuss any analytics methodology used (e.g. CRISP-DM)
- Leading from the definition of the problem, explain the data that is to be analysed.
- Explain how the data was obtained
- Detail the technical solution. Explain how the data was stored, processed and analysed. This should include details of any analytics techniques (e.g. machine learning algorithms used). This will be more of an overview compared with the technical report.
- Discuss the limitations of the project design and methods you employed.

Findings and Discussion

Findings - Describe the main results from the analysis. In addition to the written description, numerical results will usually also be presented in tabular format, either in the body of the dissertation, or in a separate appendix. This section should include a description of the results of any descriptive statistics, visualisation, machine learning models, or outputs from other analyses carried out.

Discussion of Findings – Demonstrate the significance of your arguments or results and make appropriate linkages between your findings and the wider literature.

QUEEN'S UNIVERSITY BELFAST

Conclusion and Recommendations

Conclusions– Briefly summarise what you have written from problem definition to results and discussion. Indicate the extent to which the problem has been addressed or the question has been answered and the research objectives met. Discuss how the investigation has contributed to theory and practice.

Recommendations – Consider the limitations of your research and make suggestions for future investigations. In terms of recommendations, what action should be taken based on your findings, by whom and when? Your recommendations must be consistent with, and supported by the evidence and arguments contained in your dissertation. Normally recommendations for both future research and practice should be made.

QUEEN'S UNIVERSITY BELFAST

ASSESSMENT OF RESEARCH REPORT

- Conceptual equivalents scale (appendix of handbook)
- Marking rubric (appendix of handbook)

QUEEN'S UNIVERSITY BELFAST

EXAMPLE DISSERTATION ON CANVAS

QUEEN'S UNIVERSITY BELFAST

TECHNICAL REPORT, LOG BOOK, REFLECTIVE SUMMARY

QUEEN'S
UNIVERSITY
BELFAST

TECHNICAL REPORT, LOGBOOK, REFLECTIVE DISCUSSION

- **Technical report:** The technical report should detail the key elements of the solution.
- **Logbook:** 6 entries detailing work carried out and why (around 100 words each)
- **Reflective Discussion:** Personal and professional development during the dissertation.

QUEEN'S UNIVERSITY BELFAST

TECHNICAL REPORT

Technical report: The technical report should detail the key elements of the solution. This section should not duplicate the methodology section of the research report. Rather you should focus more on the technical components, for example, why certain technical choices were made around choice of coding language and packages. What alternatives were considered when developing the technical solution. Are there any areas that could be improved, for example, to increase performance or accuracy. How does your technical solution differ from others presented in the literature. What detailed steps were taken to process the data and develop the solution? How could the solution be implemented in practice?

LOGBOOK

Logbook: Keeping a log book is a useful way of documenting the steps that were taken during the development of a solution, and the choices that were made. For this logbook, you should include six entries, which should be spaced evenly over the course of the project. These should briefly summarise the work that was carried out in the preceding time period, as well as important technical decisions that were taken and why these choices were made. Based on the log book entries, it should be possible for the reader to understand the main tasks that were carried out during the project.

REFLECTIVE DISCUSSION

Reflective discussion: The reflective discussion should focus on your personal development as a business analytics professional through carrying out the project. You should consider the key skills that you have developed, and how these will benefit you in your future career. You should also consider which elements of the project you found challenging, and where you would make improvements.

APPENDICES TO THE TECHNICAL REPORT

Appendix 1: Code that was written to develop the solution. This should be included as text in the appendix of the submission (Rather than code files).

Appendix X: Further appendices can include screenshots of the solution, as well as other technical information you feel is appropriate.

EXAMPLE TECHNICAL REPORT ON CANVAS

QUEEN'S UNIVERSITY BELFAST

DEMO HINTS AND TIPS

QUEEN'S
UNIVERSITY
BELFAST

DATA SOURCES

Dr Byron Graham
byron.graham@qub.ac.uk
@byrongraham
01/06/2026

EXAMPLE REPOSITORIES/SOURCES

- Kaggle data: www.kaggle.com
- UCI Machine Learning Repository: https://archive.ics.uci.edu/ml/index.php
- Google data search: https://datasetsearch.research.google.com/
- UCI Machine Learning Repository: https://archive.ics.uci.edu/ml/index.php
- Centre for Monitoring Indian Economy - https://www.cmie.com/
- Threads, X and other social media
- Web scraped data

QUEEN'S UNIVERSITY BELFAST

LIBRARY DATASETS (OFTEN FINANCE) / OTHER FINANCE

- Datasets available through the library: https://libguides.qub.ac.uk/business/businessdatabases
- Google finance: https://www.google.com/finance/

QUEEN'S UNIVERSITY BELFAST

OPEN (GOVERNMENT) DATA

- Open Government Data (OGD) Platform India: https://data.gov.in/
- National Statistical Office (NSO) – MOSPI: https://mospi.gov.in/
- Census of India: https://censusindia.gov.in/
- Reserve Bank of India Database on Indian Economy: https://dbie.rbi.org.in/
- National Family Health Survey (NFHS): https://dhsprogram.com/
- UN data: https://data.un.org/ or https://unstats.un.org/UNSDWebsite/

QUEEN'S UNIVERSITY BELFAST

USEFUL LISTS OF DATA SOURCES

- https://www.kdnuggets.com/datasets/index.html
- https://www.kdnuggets.com/2022/04/complete-collection-data-repositories-part-1.html
- https://www.kdnuggets.com/2022/04/complete-collection-data-repositories-part-2.html
- https://libguides.qub.ac.uk/management/businessdatabases
- https://libguides.qub.ac.uk/management/statisticsdatasets
- https://libguides.qub.ac.uk/management

QUEEN'S UNIVERSITY BELFAST

Q & A

QUEEN'S UNIVERSITY BELFAST

THANK YOU

QUEEN'S UNIVERSITY BELFAST